



IE3090

Network Programming

3rd Year, Semester I

Design Diary — IE3090 RemoteOps Assignment

Submitted to
Sri Lanka Institute of Information Technology

In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

Registration Number : IT24103381

Name : Jayarathne A W M R

Session 1 - Understanding the brief

Worked out my personalized values from IT24103381: port $7000 + 2410 = 9410$, SID = 3381 reversed = 1833, token OPS-3381, files agent_381.c / controller_381.c / Makefile_381, log remoteops_IT24103381.log and storage ./agentfiles/IT24103381/. I gave the brief to Claude to generate the first Agent, Controller and Makefile, and recorded this in my prompt log.

Session 2 - Environment and build

WSL setup failed (name-resolution error, then 'distribution already exists'), so I used my CentOS Stream 10 machine over SSH (gcc 14.4.1). I created the project folder and typed the three source files in with nano. Mistake: I typed 'cd mkdir ...' (too many arguments) and fixed it by using cd with the folder name only. The Makefile needs real Tab characters before \$(CC). Built successfully with make -f Makefile_381.

Session 3 -First run and verification

First successful run: AUTH, SYSINFO, LISTPROC, EXEC DATE / HOSTNAME, PUT and GET of testfile.txt (16 bytes, throughput printed) and QUIT, with every response ending in SID:1833. Verified personalisation: the timestamped log, the listing agentfiles/IT24103381/testfile.txt, and ss -tlnp | grep 9410 showing agent_381 listening on port 9410.

Session 4 - Stability and copying files back

My SSH session dropped (client_loop: send disconnect), which also stopped the Agent, so the Controller then printed 'Connection refused'. scp to 172.20.10.3 timed out at first and later worked from Windows PowerShell. md5sum showed controller_381.c and Makefile_381 differed from the originals (9478 vs 9477 bytes, 262 vs 352 bytes: an extra newline and two missing comment lines), which I judged harmless.

Session 5 - Remaining work

Screenshots (UDP monitor, error cases, five simultaneous Controllers, annotated code), Implementation Report, GitHub commits, prompt log, reflection, ZIP, and submission before 10:59 PM

Key decisions

- One thread per Controller connection: simple, isolated session state, meets the 5-client requirement (does not scale to thousands).
- A per-session receive buffer shared by the line reader and the byte reader, so partial lines and PUT bytes are not lost.
- A rejected upload is drained before the error is sent, so the stream stays aligned; uploads are written to a temp file and renamed.
- EXEC uses a fixed table, so client text never reaches a shell; filenames are validated to prevent path traversal.
- A single reply() adds the SID:1833 tag in one place; one UDP monitor thread per session, sending every 2 seconds.

Obstacles and fixes

- WSL install failed → used the CentOS machine over SSH.
- 'cd mkdir ...' error → used cd with the folder name only.
- Makefile needs Tabs → typed real Tabs in nano.
- SSH drop stopped the Agent, so the Controller got 'Connection refused' → restarted the Agent.
- scp timed out → ran scp from Windows PowerShell.
- md5sum mismatch → compared file sizes and found only harmless differences.